

Demonstration of PseudoJumpOn: Repetitive Step-up Jump in Virtual Reality

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Figure 1: Overview of our demo. The player makes jumping motions on a flat floor in reality (left). The jumping motions are presented as step-up jumping motions in VR and the player repeatedly makes jumps to climb up to higher platforms (middle). In the end, the system shows the trajectory of the player both in reality (orange) and VR (blue) (right).

ABSTRACT

PseudoJumpOn is a novel locomotion technique that allows step-up jumping movements using actual jumping motions on a flat floor with a common virtual reality (VR) setup. In this demo, we introduce an installation utilizing PseudoJumpOn where the player repeatedly makes step-up jumps to climb up to higher platforms. The aim is to provide players with the experience of richer vertical locomotion in VR without causing them to feel unnaturally manipulated, even though they know there are no physical steps in reality. While jumping, the system applies two types of viewpoint manipulation (i.e., gain manipulation and peak shifting) based on the user's physical jumping height and the height of the jumping target. This calculation previously assumed that the height of the jumping target was known, but we improved this by predicting it using the user's facing direction when taking off.

KEYWORDS

Jump, Locomotion, Room-scale VR

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1 INTRODUCTION

Room-scale virtual reality (VR) enables the user to have a richer and more attractive VR experience. However, since it usually requires a physical space with a flat floor and no obstacles, vertical locomotion using the user's physical movement is quite difficult. To address this problem, we previously proposed a novel vertical locomotion technique called PseudoJumpOn [Ogawa et al. 2022]. PseudoJumpOn allows the user to experience step-up jumping motions by applying two types of viewpoint-manipulation methods of gain manipulation and peak shifting to the physical jumping motions on a floor, without requiring the placement of physical steps. In contrast to existing methods such as walking up stairs or slopes, PseudoJumpOn enabled quick and casual vertical locomotion. We also showed that the technique can present step-up jumping motions in adequate reality and naturalness in the laboratory experiment.

To further demonstrate its practical applicability, we introduce an installation highlighting the use-case of PseudoJumpOn. In the demo, the players repeatedly make jumping motions onto virtual platforms and try to reach the treasure chest on the goal platform which is approximately 6-m high from the starting point. In the implementation, we address the limitation of PseudoJumpOn that

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the height of the jumping target needs to be known in advance by predicting the next jumping target (i.e., platform) using the user's facing direction when taking off. This experience will demonstrate richer locomotion, including in the vertical direction, using the user's body motion can be achieved with common VR setups.

2 DESIGN

Our aim of the demo is to provide players with the sensation of making virtual step-up jumps without causing them to feel unnaturally (forcibly) manipulated, even knowing that there are no physical steps in reality. We designed a simple virtual environment (VE) with multiple platforms, as shown in Figure 1 middle. A treasure chest is placed on the goal platform that is approximately 6-m high from the starting point, and the players' objective is to reach there by repeatedly jumping onto the platforms.

The detailed explanation of the trial is as follows. First, the player is instructed on the purpose of the experience and how to make step-up jumps (i.e., two-legged jumps as higher as possible within a safe and reasonable range, so as to actually jump onto steps). Next, the player starts the trial. The step height between each adjacent platform is low (e.g., 0.2 m) at the beginning, but it increases as the experience progresses. There are several possible routes in the series of platforms to the goal, and the player can freely choose which platform to go to next. There are also items placed on some of the platforms, and the player can make higher jumps in VR by getting them (this is achieved by increasing the value of ascending gain described below). Given our findings [Ogawa et al. 2022] that PseudoJumpOn can present an adequately realistic and natural jumping experience even in the maximum height (0.8 m) and gain (5.0) tested in the experiment, this demo presents even higher steps and larger gains (at the maximum of 1.2 m and 6.0, respectively, depending on the route the player chooses) that are also expected to be acceptable. If the player successfully reaches the goal where a treasure chest is placed, the trial is complete and some statistics (e.g., number of jumps, the average jumped height) are shown. If the player could not make a jump because of the insufficient jump distance or height, the game will be over and she/he falls to the ground. After the trial is completed or over, the system presents the trajectory of the player both in reality and VR, and she/he can see the difference in movement between them to recognize how our system enhances the jump experience (Figure 1 right).

3 IMPLEMENTATION

Our implementation is mostly in line with that of the original PseudoJumpOn study [Ogawa et al. 2022]. We chose Oculus Quest 2 as a head-mounted display because it can perform 6-DoF tracking by itself without interference by cables. For the player's safety, we also place a soft floor mat in case of their falls. For the detection of jumping motions, we implemented the system so that it considers the user is jumping if the user's head position is higher than a certain threshold $\theta_{takeoff}$ (0.11 m, as in the original work). For the viewpoint manipulation during jumping, PseudoJumpOn applies two types of algorithms, gain manipulation and peak shifting, as shown in Figure 2. We describe the detailed implementation below.

Gain manipulation applies different gains between ascent and descent and creates a difference in height between them. Following the original work [Ogawa et al. 2022], we employed a method

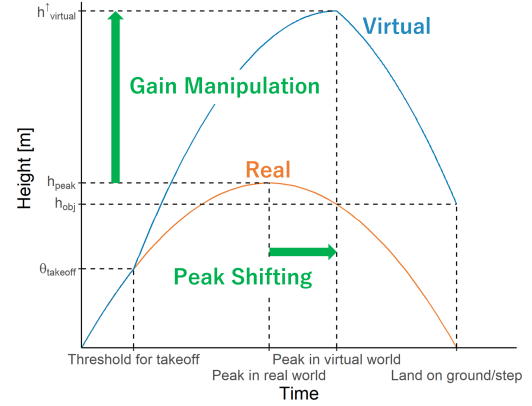


Figure 2: Principle of PseudoJumpOn. The landing height is manipulated by gain manipulation and peak shifting.

where the ascending gain is fixed and the descending gain is calculated dynamically. The ascending gain is 1 at the beginning of the experience and it increases when the player gets an item during the experience, enabling the player to make higher jumps. The descending gain $g_v^\downarrow$ is calculated by the following equation.

$$g_v^\downarrow = \frac{h_{virtual}^\uparrow - h_{obj}}{h_{remain}} \quad (1)$$

Here, $h_{virtual}^\uparrow$ is the player's jumped height in VR, h_{obj} is the height of the platform, and h_{remain} is the player's height in reality when reaching the highest point in VR. If $g_v^\downarrow$ is calculated as a negative value, the system judges as the user could not reach the platform.

Peak shifting manipulates the user's viewpoint so that the ascent duration is longer than the descent and creates a difference in time between them. The variable peak shift ps indicates how much the peak point is delayed, and it is calculated by the following equation using the ascent duration in reality $t_{physical}^\uparrow$ and in VR $t_{virtual}^\uparrow$.

$$ps = \frac{t_{virtual}^\uparrow - t_{physical}^\uparrow}{t_{physical}^\uparrow} \quad (2)$$

In our demo, we used a fixed value of 0.5 (halfway between 0 and 1.0) as the peak shift because we previously found that peak shifting does not significantly affect the user experience [Ogawa et al. 2022].

We additionally employed an algorithm that predicts the next jumping target by the user's facing direction, considering the case that there is more than one platform that the user could reach from the current position. We implemented it so that it selects the nearest target (i.e., platform) that is in the direction of the user facing and within the range of a certain threshold (we set this as 2 m from the preliminary tests) as the next target.

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